Questions

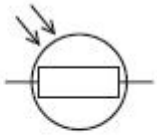
Q1.

Some questions must be answered with a cross in a box (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which of these shows the correct circuit symbol for a thermistor?

(1)

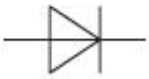
☐ **A**



☐ **B**

☐ **C**

☐ **D**



(Total for question = 1 mark)

Q2.

Figures 21 and 22 show different voltages that can be applied across a wire.

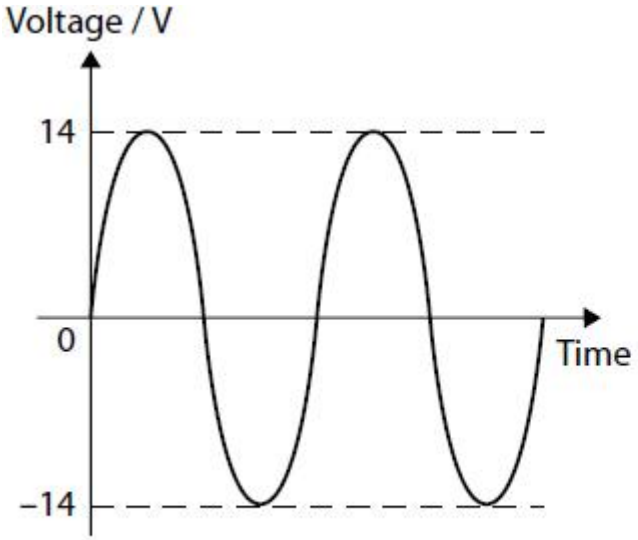


Figure 21

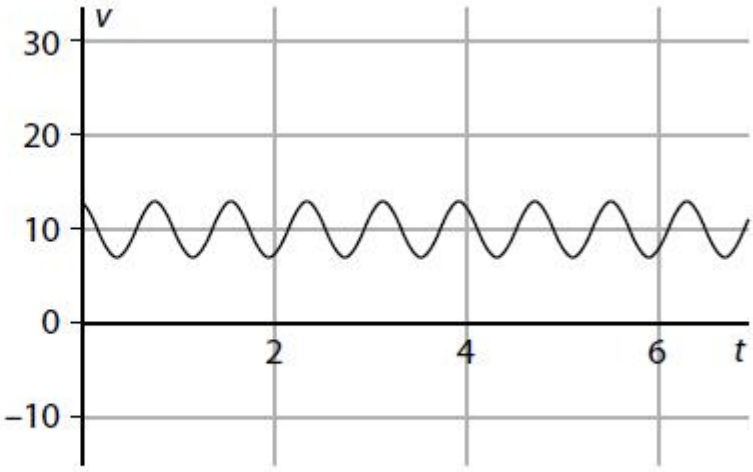


Figure 22

Explain which of the voltages in Figures 21 and 22 cause an a.c. current in the wire.

(2)

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(Total for question = 2 marks)

Q3.

The photograph shows a step-down transformer.

(a) Explain why step-down transformers are used in the transmission of electricity in the National Grid.

(2)

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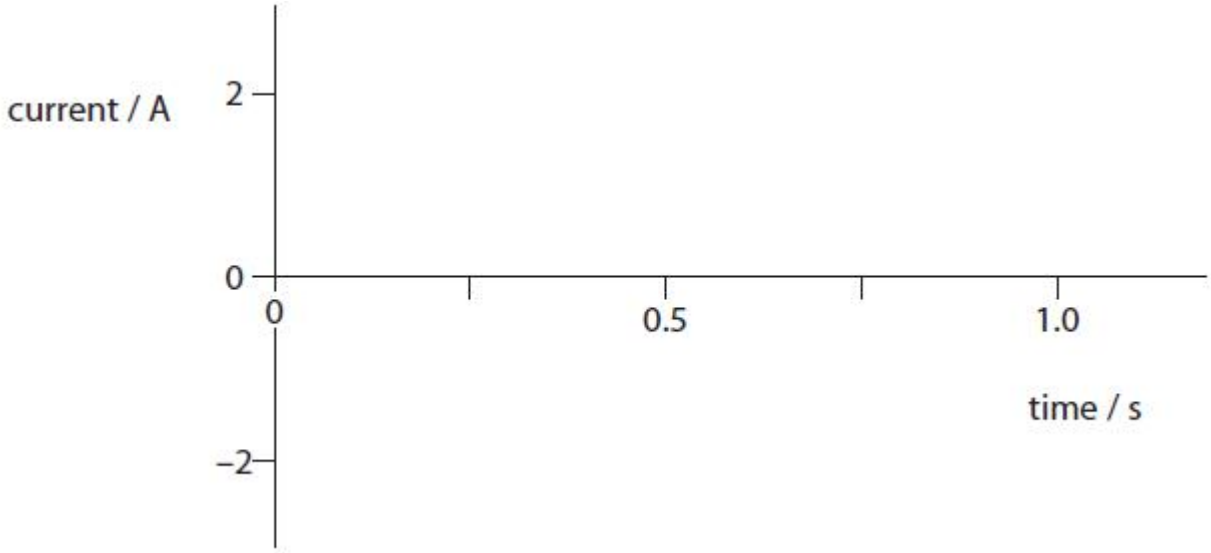
(b) Transformers need alternating current to work properly.
Sketch a graph of an alternating current with a frequency of 2 Hz.

(2)



(c) A transformer has 2400 turns on the primary coil and 100 turns on the secondary coil.

Calculate the secondary voltage if the primary voltage is 12 V.

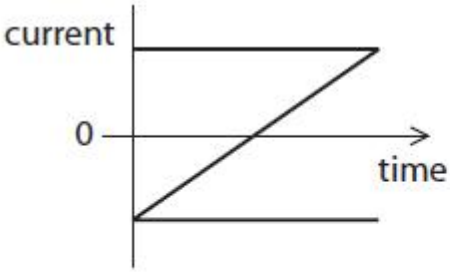
(3)

secondary voltage =..... V

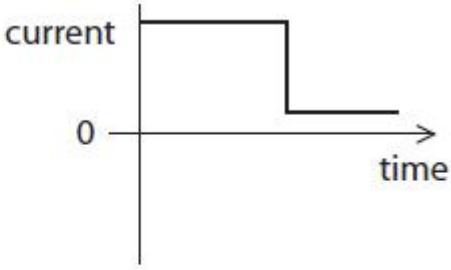
(d) Opening the switch in a circuit produces the opposite magnetic effect to closing the switch.
A scientist connected a switch, a fixed resistor and a battery to the primary coil of a step-up transformer.
The scientist also connected a fixed resistor across the secondary coil.
The scientist switched the circuit on and then later switched it off.

Which of these best represents the current in the secondary coil?

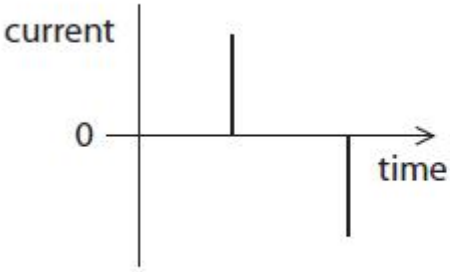
(1)



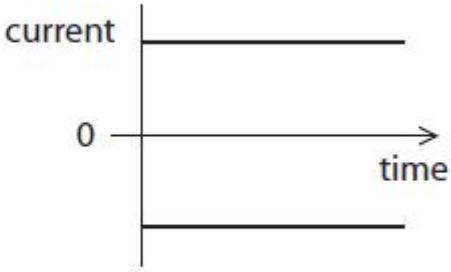
☒ **A**



☒ **B**



☒ **C**



☒ **D**

(Total for Question = 8 marks)

Q4.

(i) Describe **one** situation where separation of electric charge can create a spark.

(2)

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(ii) In a spark, the total charge of 0.22 μC (microcoulombs) flows in 2 ms (milliseconds).
Calculate the average current in that time.

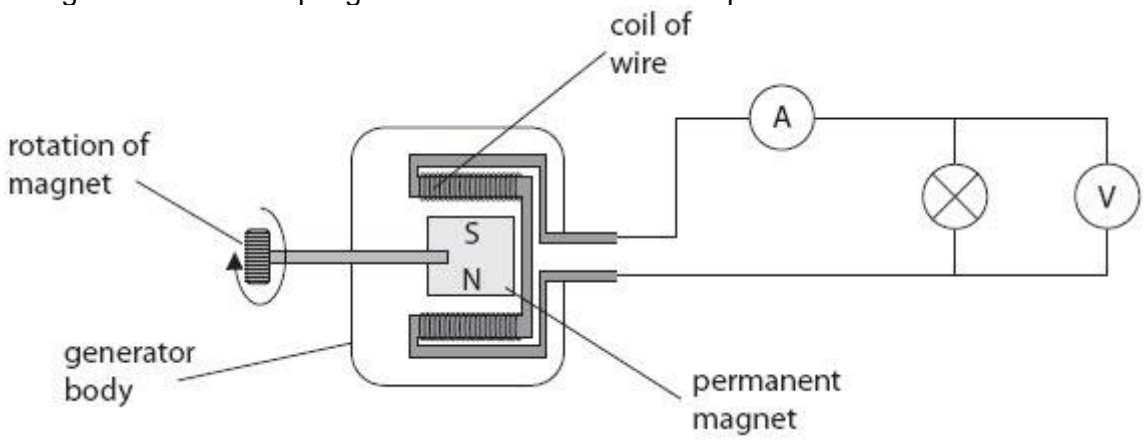
(4)

average current = ... A

(Total for question = 6 marks)

Q5.

(a) The diagram shows a simple generator connected to a lamp.



The magnet is made to spin at a steady speed.
The ammeter gives a reading of 1.5 A.
The voltmeter gives a reading of 6 V.
(i) Calculate the output power of the generator.

(2)

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(ii) State two changes to the design of the generator that would give a larger output power for the same speed of rotation.

(2)

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(iii) This generator supplies an alternating current (AC) to the lamp.
Other types of generators supply a direct current (DC).
Describe the difference between charge movement in a direct current and in an alternating current.

(2)

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*(b) The first public power station was built in the centre of New York.

It used generators to supply direct current at 110 V. The cables had to go underground and they could only supply
nearby shops and offices.

The electricity was mainly used for electric light.

The development of alternating current generators led to major changes in the way electricity is transmitted and used.

Compare the modern National Grid system with the early system in New York.

(6)

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(Total for Question is 12 marks)

Q6.

Figure 16 shows a cardboard tube with a wire coming out from each end.

Figure 16

There are two 10 ohm resistors inside the cardboard tube.

A potential difference of 6.0 V is connected between P and Q.

There is a current of 1.2 A in the wires.

Deduce how the resistors have been arranged inside the cardboard tube.

(3)

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(Total for question = 3 marks)

Q7.

A car headlamp has a power rating of 55 W when the current in the headlamp is 4.4 A.

(i) State the equation relating power, current and resistance.

(1)

(ii) Calculate the resistance of the headlamp.

(3)

resistance = ... Ω

(Total for question = 4 marks)

Q8.

State the frequency of the supply of electricity to homes in the UK (United Kingdom).

(1)

frequency = ... Hz

(Total for question = 1 mark)

Mark Scheme

Q1.

Question number	Answer	Mark
	☒ C Only this is the correct symbol for a thermistor	(1) AO1.1

Q2.

Question number	Answer	Mark
	An explanation that combines identification – understanding (1 mark) and reasoning/justification – understanding (1 mark): • if the voltage changes sign, then the current is changing direction • so in Figure 21 the current is a.c. as the voltage is changing sign and in Figure 22 the current is d.c. as the voltage is always positive	(2)

Q3.

Question Number	Answer	Acceptable answers	Mark
(a)	An explanation linking two from		
	MP1 (so that they) decrease the (high) voltages (1)	**stepping down voltage reducing from {high/eg 200 000 V} to {low /e.g.230 V} voltage**	
	MP2 high voltages used for efficiency/energy saving (1)	**low current used for efficiency/ energy saving**	
	MP3 (step-down transformers) used {near / for} {homes / factories/appliances} (1)		
	MP4 (so that it is) safer (1)	less risk of electrocution high voltages are dangerous	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b)	one line / curve above **and** below x-axis (1)		
	two complete cycles in the 1.0 s (1)	one complete cycle in 0.5 s	**(2)**

Question Number	Answer	Acceptable answers	Mark
(c)	Transposition (1) $V_s = V_p \times n_s/n_p$ Substitution (1) $(V_s =) \frac{12 \times 100}{2400}$ Evaluation (1) 0.5 (V)	Substitution and transposition in either order i.e. if $\frac{12 \times 100}{2400}$ is seen this scores 2 If they sub Vp, Np and Ns correctly, ignore anything for Vs even a blank Calculation may be done using turns ratio Correct answer no working = full marks answer (no working) with POT error =2 (eg 5 or 0.05) Ignore powers of 10 until evaluation	**(3)**

Question Number	Answer	Acceptable answers	**Mark**
(d)	C		**(1)**

Q4.

Question number	**Answer**	**Additional guidance**	**Mark**
(i)	An answer that combines the following points of understanding to provide a logical description: the situation which caused the charge separation (1) where the spark travelled {from/to} (1)	examples: when refuelling, spark between end of {fuel/pipe} and vehicle =2 spark {between/from /to} person comb/clothes/metal handle and, when combing hair/removing clothing/opening door = 2 lightning flash, between cloud and cloud/plane/ground, =2 ignore between plug and socket/jump leads	**(2)**

Question number	Answer	Additional guidance	Mark
(ii)	unit conversion (1) 0.22 μC = 0.22×10^{-6} C and 2 ms = 2×10^{-3} s substitution (1) 0.22×10^{-6} = current x 2×10^{-3} s rearrangement (1) current = $0.22 \times 10^{-6} / 2 \times 10^{-3}$ evaluation (1) 1.1×10^{-4} (A)	substitution and re-arrangement in either order both needed award full marks for correct answer without working power of ten error only loses one mark, if the rest is correct	(4)

Q5.

	Answer	Acceptable answers	Mark
(ai)	Substitution (1) 1.5 × 6 Evaluation (1) 9 (W) Ignore any unit given by candidate.	Power of 10 error max 1 mark Give full marks for correct answer with no working shown	(2)
(aii)	• More turns on the coil (1) • More powerful/stronger magnet(s) (1)	Wrap coils on iron (core/former)/ more coils/twists/loops. Bigger coil is insufficient. More magnets. Bigger/larger magnet is insufficient. Ignore increase speed of rotation	(2)
(aiii)	A description including • in one direction only for DC (1) • reversing direction for AC (1)	'DC goes straight' is insufficient AC switches/changes direction OR moves to and fro 'AC goes different ways' is insufficient. Diagram with labelled arrows could get 2 marks.	(2)

		Indicative Content	Mark

QWC	***(b)**	A comparison including some of the following ideas • Transformers can be used or voltages/currents can be changed/transformed • AC (can transmit) at lower current/high(er) voltage • National Grid is (usually) over ground (DC cables (were) underground) • Less energy lost in transmission • National Grid system can supply to customers further away • Possible to create a grid linking power stations • More flexibility in voltage for consumer • Consumer can draw large(r) current • More flexibility in power drawn • Great(er) range of devices can be powered Ignore methods of electricity production	**(6)** **Exp**
Level	**0**	No rewardable content	
1	**1 - 2**	• a limited (maybe implied) comparison giving one fact e.g: AC can be at high(er) voltage OR the National Grid can supply houses not close to a power station/ further (away/than the New York system.) • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• a simple comparison including two ideas which may be linked or not eg Nat. Grid can supply whole country and can be used for more appliances (than just lighting). e.g: AC can be transmitted further (than DC) (because it) wastes less energy • the answer communicates ideas	

		showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy
3	**5 - 6**	• A detailed comparison including at least three ideas, with at least one direct link between two of them. • e.g. AC can be transmitted further (than DC) because AC can be transformed to lower current/high(er) voltages. OR AC can be transformed to lower current/high(er) voltages. Greater range of devices used. • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors

Total for Question = 12 mark

Q6.

Question Number:	Answer	Additional Guidance	Mark
	an explanation linking:	allow alternative arguments such as if resistors had been in series, then...	**(3)** AO 3 2a AO 3 2b
	relevant calculation (1) R (between P and Q) = $\frac{6}{1.2}$ = 5Ω	I = $\frac{6}{20}$ = 0.3A V (between P and Q) = 1.2 x 10 = 12V	
	reasoning / interpretation of result (1) this is less than {a single resistor / two resistors in series}	current is more (than 0.3A) total p.d. is less than 12 V	
	conclusion (1) resistors must be connected <u>in parallel</u>		

Q7.

Question number	Answer	Mark
(i)	either power = (current)^2 × resistance OR $P = I^2 \times R$	**(1)**

Question number	Answer	Additional guidance	Mark
(ii)	substitution into $P = I^2 \times R$ (1) $55 = 4.4^2 \times R$ rearrangement (1) $R = P \div I^2$ evaluation (1) 2.8 (Ω)	Substitution and re-arrangement in either order $R = 55 \div 4.4^2$ for 2 marks allow answers that round to 2.8 award full marks for correct answer without working allow alternative route $V = P \div I = 55 \div 4.4$ then $R = V \div I = 12.5 \div 4.4$	(3)

Q8.

Question Number	Answer	Additional guidance	Mark
	50(Hz) (1)		(1) AO1.1